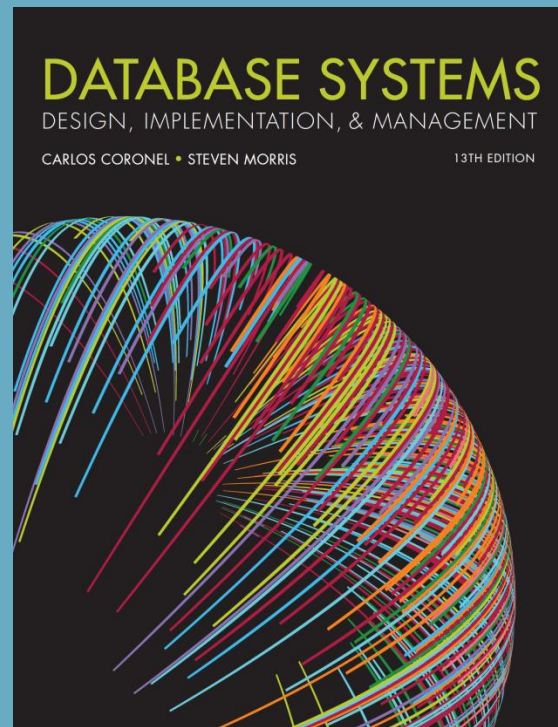




Chapter 1

Database Systems



Learning Objectives

- After completing this chapter, you will be able to:
 - Define the difference between data and information
 - Describe what a database is, various types, and why they are valuable assets for decision making
 - Explain the importance of database design
 - See how modern databases evolved from file systems
 - Understand flaws in file system data management
 - Outline the main components of the database system
 - Describe the main functions of a database management system (DBMS)



Why Databases?

- Characteristics of data in today's world
 - Ubiquitous (i.e., abundant, global, and everywhere)
 - Pervasive (i.e., unescapable, prevalent, and persistent)
- Databases make data persistent and shareable in a secure way
 - Specialized structures that allow computer-based systems to store, manage, and retrieve data very quickly

FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES

A Day In Susan's Life

See how many databases she interacts with each day

*Before leaving for work,
Susan checks her
Facebook and
Twitter accounts*



*On her lunch break,
she picks up her
prescription at the
pharmacy*



*After work, Susan
goes to the grocery
store*



*At night, she plans for a trip
and buys airline tickets and
hotel reservations online*



*Then she makes a few
online purchases*



Where is the data about the
friends and groups stored?

Where are the “likes” stored
and what would they be
used for?



Where is the pharmacy
inventory data stored?

What data about each
product will be in the
inventory data?

What data is kept about
each customer and where
is it stored?



Where is the product
data stored?

Is the product quantity in
stock updated at checkout?

Does she pay with a credit
card?



Where does the online
travel website get the
airline and hotel data from?

What customer data would
be kept by the website?

Where would the customer
data be stored?



Where are the product
and stock data stored?

Where does the system get
the data to generate product
“recommendations” to the
customer?

Where would credit card
information be stored?





Data versus Information

- Data consists of raw facts
 - Not yet processed to reveal meaning to the end user
 - Building blocks of information
- Information results from processing raw data to reveal meaning
 - Requires context
 - Bedrock of knowledge
 - Should be accurate, relevant, and timely

<http://www.knowledge-management-tools.net/knowledge-information-data.html>

<https://www.youtube.com/watch?v=QsP5WGV0aQc>

<https://www.highperformancelearning.com/data-information-knowledge-and-wisdom/>

<https://www.systems-thinking.org/dikw/dikw.htm>



FIGURE 1.2 TRANSFORMING RAW DATA INTO INFORMATION

a) Data entry screen

Middle Tennessee State University

You are viewing the College console.

Select Member

Home Reports Activity Aggregations Summaries Maintenance Mission & Accreditation Managers Subsites & Tools Calendar Directory Logs Settings Home Sign Out

Jennings A. Jones College of Business

Home Manage Members Add Faculty Form

DO NOT append School ID (MT) to Member ID

Member ID* (7 alphanumeric) (Password will be initially set to be the same as Member ID)

First name or initial*

Middle name/initial*

Last name*

☐ Chair/lead ☐ Inactive ☐ Opens chair for evaluation

Department*

Area*

Email*

Hire Term*

Member Default Status: Changing the status here changes only the default that is pulled into the uploaded teaching schedules. To change the historical status of members and to see your changes reflected in the various reports, edit the teaching schedules themselves.

Involvement* ☐ Participating ☐ Supporting

Qualification*

☐ Participates in the governance of the school

☐ Considered to be a long term member

High Degree*

Year Awarded*

Rank* Assistant Professor

b) Raw data

ID	Last Name	Mid Name	First Name	Dept Code	Office	Email	Rank	Hire Year	Degree
1	Washington	A.	George	MGMT	N135	g.washington@mtsu.edu	Professor	2011	Ph.D.
2	Adams		John	FIN	N113	j.adams@mtsu.edu	Professor	1984	Ph.D.
3	Jefferson	L.	Thomas	ECOM		l.jefferson@mtsu.edu	Instructor	1994	M.B.A.
4	Madison	D.	James	FIN	N126	j.madison@mtsu.edu	Associate Professor	1994	Ph.D.
5	Monroe	N.	James	ACCT	N411	j.monroe@mtsu.edu	Associate Professor	1995	Ph.D.
6	Adams	G.	John	ACCT	N418	j.adams@mtsu.edu	Associate Professor	1999	Ph.D.
7	Jackson	C.	Andrew	ECOM	N103	a.jackson@mtsu.edu	Associate Professor	1999	Ph.D.
8	Van Buren	T.	Math	FIN	N106	m.vanburen@mtsu.edu	Professor	1998	Ph.D.
9	Harrison	R.	William	MKTG	N118	w.harrison@mtsu.edu	Professor	1994	Ph.D.
10	Tyler	M.	John	MGMT		j.tyler@mtsu.edu	Associate Professor	2000	E.D.
11	Fink		Cheryl	MKTG	N140	c.fink@mtsu.edu	Associate Professor	2002	Ph.D.
12	Taylor	G.	Zachary	ACCT	N414	z.taylor@mtsu.edu	Associate Professor	1996	Ph.D.
13	Fillmore		Malcolm	JOB	N119	m.fillmore@mtsu.edu	Professor	1992	Ph.D.
14	Pierre	A.	Franklin	MKTG	N159	f.pierre@mtsu.edu	Instructor	2005	M.B.A.
15	Buchanan	T.	James	MGMT	N146	j.buchanan@mtsu.edu	Associate Professor	1986	D.B.A.
16	Jacobs	W.	Larry	MGMT	N158	l.jacobs@mtsu.edu	Associate Professor	1986	Ph.D.
17	Johnson		Andrew	ISYS	N166	a.johnson@mtsu.edu	Professor	1987	Ph.D.
18	Goss		Kelly	MKTG	N128	k.goss@mtsu.edu	Associate Professor	1989	D.B.A.
19	Rutherford		Hayes	ACCT	N408	h.rutherford@mtsu.edu	Professor	1992	Ph.D.
20	Orfield	T.	Dennis	ACCT		d.orfield@mtsu.edu	Associate Professor	2010	Ph.D.
21	Arthur		Emily	ACCT	N413	e.arthur@mtsu.edu	Associate Professor	2007	J.D.
22	Cleveland	G.	Robert	ACCT	N401	r.cleveland@mtsu.edu	Associate Professor	1997	Ph.D.
23	Harrison	X.	Patrick	BULA	N406	p.harrison@mtsu.edu	Associate Professor	2001	J.D.
24	McKinley	B.	Priscilla	ISYS	N161	p.mckinley@mtsu.edu	Adjunct	1994	M.S.
25	Roosevelt	F.	Hilary	MGMT	N104	h.roosevelt@mtsu.edu	Associate Professor	2002	Ph.D.
26	Wilson		Laura	ECOM	N448	l.wilson@mtsu.edu	Professor	1992	Ph.D.
27	Harding		Wanda	MKTG	N114	w.harding@mtsu.edu	Professor	1984	E.D.
28	Coledge		Darin	ECOM	N116	d.coledge@mtsu.edu	Professor	1976	Ph.D.
29	Holzer		Lisa	MGMT		l.holzer@mtsu.edu	Adjunct	1976	M.B.A.
30	Tanner	B.	Betty	ACCT	N416	b.tanner@mtsu.edu	Professor	1971	E.D.
31	Johnson		Robert	ECOM	N140	r.johnson@mtsu.edu	Professor	2001	Ph.D.

c) Information in summary format

Rank	COUNT	%/INFS	TOT/COL	%/COL. TOT.	%/COL. FAC.
Adjunct	5	20.00%	23	21.74%	3.27%
Assistant Professor	2	8.00%	28	7.14%	1.31%
Associate Professor	9	36.00%	37	24.32%	5.88%
Instructor	2	8.00%	18	11.11%	1.31%
Professor	7	28.00%	47	14.89%	4.58%

d) Information in graphical format

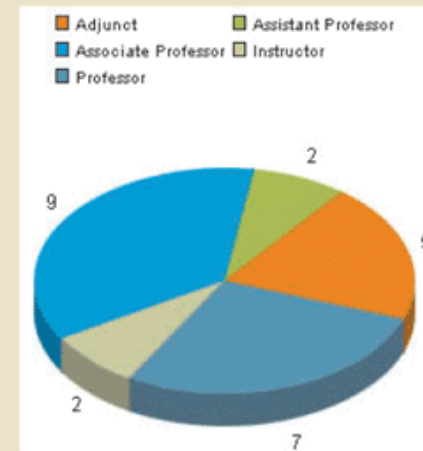
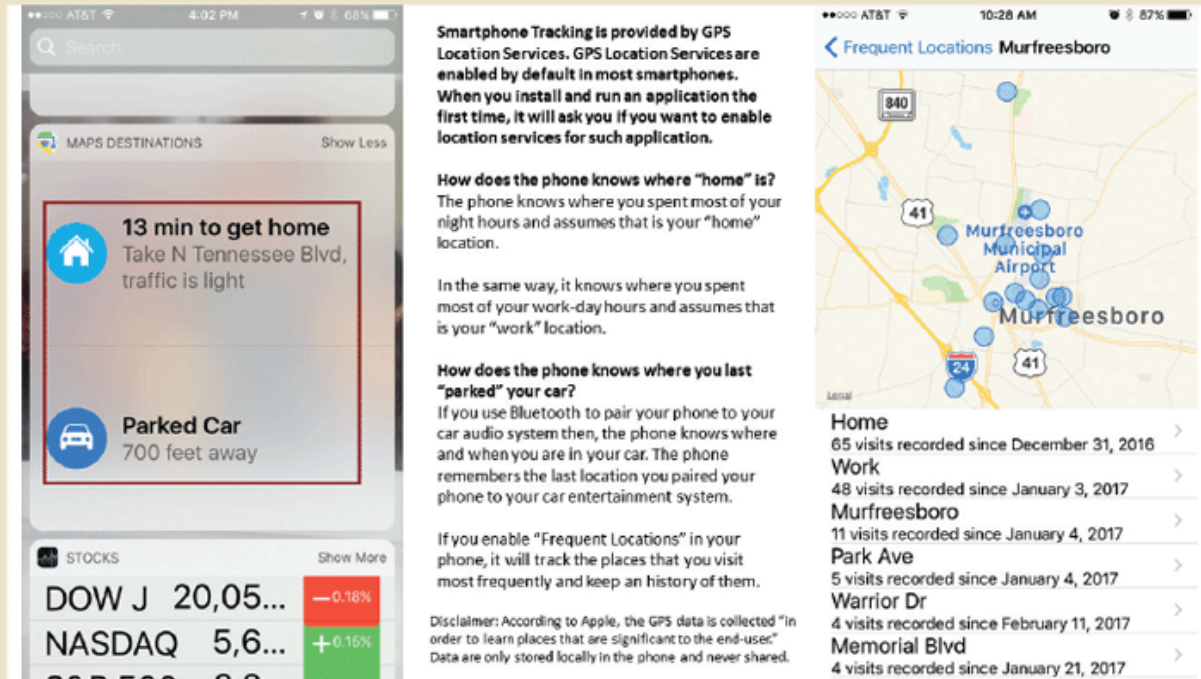




FIGURE 1.3 SMARTPHONE TRACKING





Breakout Room

- From Figure 1.1
 - List all the important data that should be collected to relate with the Smartphone Tracking context
 - Show the data in the form of Tabular based on your discussion

Column_name1	Column_name2	Column_name3	...	
Data	Data			
Data	...			
Data				

- Show some example of marketing strategies that business can use to persuade Susan to buy their products.



Introducing the Database

- Shared, integrated computer structure that stores data
 - End-user data: raw facts of interest to end user
 - Metadata: data about data, through which the end-user data is integrated and managed
 - Describes data characteristics and relationships
- Database management system (DBMS)
 - Collection of programs
 - Manages the database structure
 - Controls access to data stored in the database

<https://www.learncomputerscienceonline.com/database-management-system/>



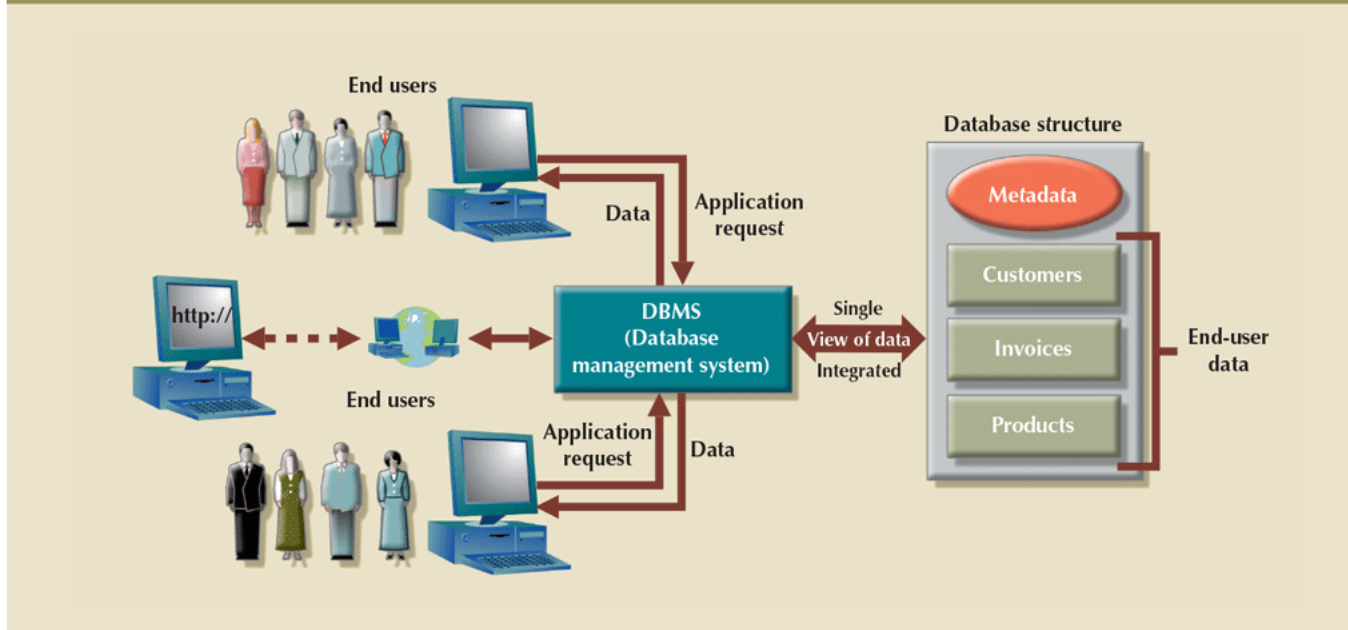
Role and Advantages of the DBMS (1 of 2)

- Database management system (DBMS): intermediary between the user and the database
 - Enables data to be shared
 - Presents the end user with an integrated view of data
 - Provides more efficient and effective data management
 - Improves sharing, security, integration, access, decision-making, productivity, etc.



Role and Advantages of the DBMS (2 of 2)

FIGURE 1.4 THE DBMS MANAGES THE INTERACTION BETWEEN THE END USER AND THE DATABASE





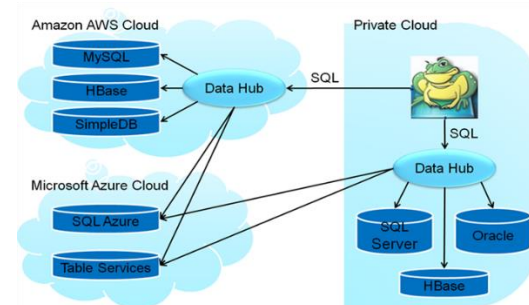
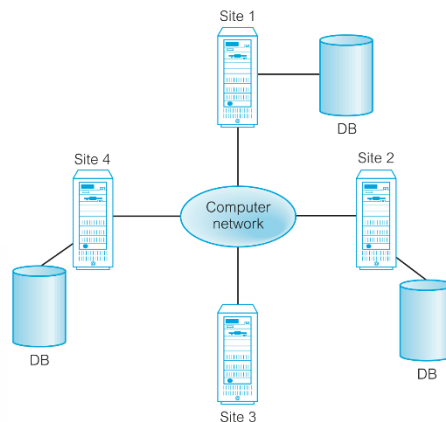
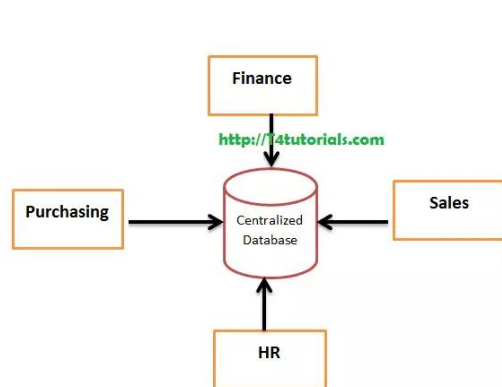
Types of Databases (1 of 5)

- Single-user database: supports one user at a time
 - Desktop database: single-user database on a personal computer
- Multiuser database: supports multiple users at the same time
 - Workgroup databases: supports a small number of users or a specific department
 - Enterprise database: supports many users across many departments



Types of Databases (2 of 5)

- Classification by location
 - Centralized database: data located at a single site
 - Distributed database: data distributed across different sites
 - Cloud database: created and maintained using cloud data services that provide defined performance measures for the database



<https://www.ariseserver.com/cloud-database/>

<https://sungsoo.github.io/2014/05/17/distributed-database-management-system-introduction.html>



Types of Databases (3 of 5)

- Classification by data type
 - General-purpose database: contains a wide variety of data used in multiple disciplines
 - Discipline-specific database: contains data focused on specific subject areas
 - Operational database: designed to support a company's day-to-day operations



Types of Databases (4 of 5)

- Analytical database: stores historical data and business metrics used exclusively for tactical or strategic decision making
 - Data warehouse: stores data in a format optimized for decision support
 - Online analytical processing (OLAP): tools for retrieving, processing, and modeling data from the data warehouse
 - Business intelligence: captures and processes business data to generate information that support decision making



https://www.123rf.com/photo_91962706_stock-vector-data-warehouse-icon-logo-design-vector-illustration-technology-solution-trend-concept-design-.html

shutterstock.com • 1260228499

<https://businessimpactinc.com/blog/predictive-analytics-what-it-is-how-it-works/>



Types of Databases (5 of 5)

- Databases can be classified to reflect the degree to which the data is structured
 - Unstructured data exists in its original (raw) state
 - Structured data results from formatting
 - Structure is applied based on type of processing to be performed
 - Semistructured data: processed to some extent
- Extensible Markup Language (XML)
 - Represents data elements in textual format



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  <Book ISBN="ISBN-0-13-035300-0" Price="$65" Edition="2nd">
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        <Last_Name>Ullman</Last_Name>
      </Author>
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    <Remark>
      Amazon.com says: Buy this book bundled with "A First Course,"
      it's a great deal!
    </Remark>
  </Book>
</Bookstore>
```



Why Database Design Is Important

- Focuses on design of database structure that will be used to store and manage end-user data
 - Well-designed database: facilitates data management and generates accurate and valuable information
 - Poorly designed database: causes difficult-to-trace errors that may lead to poor decision making



FIGURE 1.5 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN

Why are there blanks in rows 9 and 10?

How to produce an alphabetical listing of employees?

How to count how many employees are certified in Basic Database Manipulation?

Is Basic Database Manipulation the same as Basic DB Manipulation?

What if an employee acquires a fourth certification?
Do we add another column?

ID	ENum	Name	Title	HireDate	Skill1	Skill1Date	Skill2	Skill2Date	Skill3	Skill3Date
1	02345	Brian Oates	DBA	2/14/1997	Basic Database Management	2/14/2004	Advanced Database Management	2/14/2007	Basic Web Design	8/9/2005
2	08273	Marco Bienz	Analyst	7/28/2008	Basic Web Design	3/8/2011	Advance Process Modeling	8/19/2014		
3	06234	Jasmine Patel	Programmer	8/10/2007	Basic Web Design	8/10/2009	Advanced C# programming	8/10/2009	Basic DB manipulation	1/29/2014
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2004	Advanced Spreadsheets	6/20/2013				
5	13567	Almond, Robert	Analyst	9/30/2014	Basic Process Modeling	9/30/2016	Basic Database Design	5/23/2017		
6	10282	Richardson, Amanda	Clerk	4/11/2013						
7	09382	Susan Mathis	Database Programmer	8/2/2012	Basic DB Design	8/2/2014	Basic Database Manipulation	8/2/2014	Advanced DB Manipulation	5/1/2015
8	14311	Duong, Lee	Programmer	9/1/2016	Basic Web Design	9/1/2018				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2012	Advanced Spreadsheets	5/16/2015	Basic Web Design	5/16/2015		
12	13383	Raymond F. Matthews	Programmer	3/12/2014	Basic C# Programming	3/12/2016				
13	09283	Chavez, Juan	Clerk	7/4/2012						
14	04893	Patricia Richards	DBA	6/11/2006	Advanced Database Management	6/11/2008	Advanced Database Manipulation	9/20/2014		
15	13932	Lee, Megan	Programmer	9/29/2015						



Evolution of File System Data Processing (1 of 3)

- Manual file systems
 - Accomplished through a system of file folders and filing cabinets
- Computerized file systems
 - Data processing (DP) specialist created a computer-based system to track data and produce required reports
- File system redux: modern end-user productivity tools
 - Includes spreadsheet programs such as Microsoft Excel



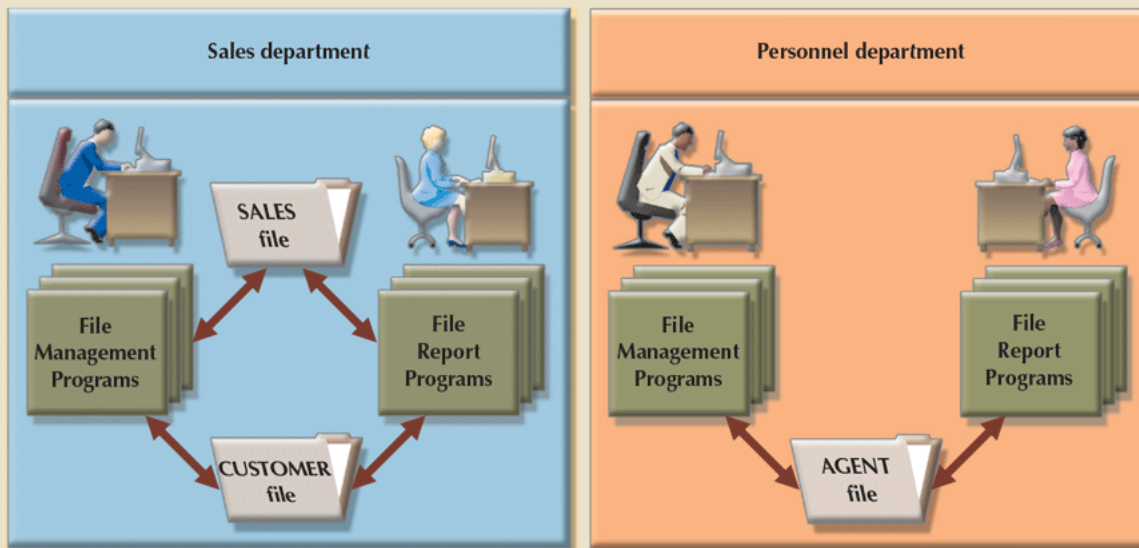
Evolution of File System Data Processing (2 of 3)

Table 1.2	Basic File Terminology
TERM	DEFINITION
Data	Raw facts, such as a telephone number, a birth date, a customer name, and a year-to-date (YTD) sales value. Data has little meaning unless it has been organized in some logical manner.
Field	A character or group of characters (alphabetic or numeric) that has a specific meaning. A field is used to define and store data.
Record	A logically connected set of one or more fields that describes a person, place, or thing. For example, the fields that constitute a record for a customer might consist of the customer's name, address, phone number, date of birth, credit limit, and unpaid balance.
File	A collection of related records. For example, a file might contain data about the students currently enrolled at Gigantic University.



Evolution of File System Data Processing (3 of 3)

FIGURE 1.9 A SIMPLE FILE SYSTEM





Problems with File System Data Processing

- Problems with file systems challenge the types of information that can be created from data as well as information accuracy
 - Lengthy development times
 - Difficulty of getting quick answers
 - Complex system administration
 - Lack of security and limited data sharing
 - Extensive programming



Structural and Data Dependence (1 of 2)

- Structural dependence
 - Access to a file is dependent on its own structure
 - All file system programs are modified to conform to a new file structure
- Structural independence
 - File structure is changed without affecting the application's ability to access the data



Structural and Data Dependence (2 of 2)

- Data dependence
 - Data access changes when data storage characteristics change
- Data independence
 - Data storage characteristics are changed without affecting the program's ability to access the data
- Practical significance of data dependence is the difference between logical and physical format



Data Redundancy (1 of 2)

- Unnecessarily storing the same data at different places
 - Islands of information (i.e., scattered data locations)
 - Increases the probability of having different versions of the same data



Data Redundancy (2 of 2)

- Possible results of uncontrolled data redundancy
 - Poor data security
 - Data inconsistency
 - Data-entry errors
 - Data integrity problems



Data Anomalies

- Develop when not all of the required changes in the redundant data are made successfully
 - Update anomalies
 - Insertion anomalies
 - Deletion anomalies

FIGURE 1.6 EMPLOYEE SKILLS CERTIFICATION IN A GOOD DESIGN



Table name: EMPLOYEE

Employee_ID	Employee_FName	Employee_LName	Employee_HireDate	Employee_Title
02345	Brian	Oates	2/14/1999	DBA
03373	Franklin	Johnson	3/15/2006	Purchasing Agent
04893	Patricia	Richards	6/11/2008	DBA
06234	Jasmine	Patel	8/10/2009	Programmer
08273	Marco	Bienz	7/28/2010	Analyst
09002	Wade	Gaither	5/20/2014	Clerk
09283	Juan	Chavez	7/4/2014	Clerk
09382	Susan	Mathis	8/2/2014	Database Programmer
10282	Amanda	Richardson	4/11/2015	Clerk
13383	Raymond	Matthews	3/12/2016	Programmer
13567	Robert	Almond	9/30/2016	Analyst
13932	Megan	Lee	9/29/2017	Programmer
14311	Lee	Duong	9/1/2018	Programmer

Database name: Ch01_Text

Table name: CERTIFIED

Employee_ID	Skill_ID	Certified_Date
02345	100	2/14/2004
02345	110	8/9/2005
02345	180	2/14/2007
03373	120	6/20/2013
04893	180	6/11/2008
04893	220	9/20/2014
06234	110	8/10/2009
06234	200	8/10/2009
06234	210	1/29/2014
08273	110	3/8/2011
08273	190	8/19/2014
09002	110	5/16/2015
09002	120	5/16/2015
09382	140	8/2/2014
09382	210	8/2/2014
09382	220	5/1/2015
13383	170	3/12/2016
13567	130	9/30/2016
13567	140	5/23/2017
14311	110	9/1/2018

Table name: SKILL

Skill_ID	Skill_Name	Skill_Description
100	Basic Database Management	Create and manage database user accounts.
110	Basic Web Design	Create and maintain HTML and CSS documents.
120	Advanced Spreadsheets	Use of advanced functions, user-defined functions, and macroing.
130	Basic Process Modeling	Create core business process models using standard libraries.
140	Basic Database Design	Create simple data models.
150	Master Database Programming	Create integrated trigger and procedure packages for a distributed environment.
160	Basic Spreadsheets	Create single tab worksheets with basic formulas
170	Basic C# Programming	Create single-tier data aware modules.
180	Advanced Database Management	Manage Database Server Clusters.
190	Advance Process Modeling	Evaluate and Redesign cross-functional internal and external business processes.
200	Advanced C# Programming	Create multi-tier applications using multi-threading
210	Basic Database Manipulation	Create simple data retrieval and manipulation statements in SQL.
220	Advanced Database Manipulation	Use of advanced data manipulation methods for multi-table inserts, set operations, and correlated subqueries.

FIGURE 1.7 CONTENTS OF THE CUSTOMER FILE

Database name: Ch01_Text

C_NAME	C_PHONE	C_ADDRESS	C_ZIP	A_NAME	A_PHONE	TP	AMT	REN
Alfred A. Ramas	615-844-2573	218 Fork Rd., Babs, TN	36123	Leah F. Hahn	615-882-1244	T1	100.00	05-Apr-2018
Leona K. Dunne	713-894-1238	Box 12A, Fox, KY	25246	Alex B. Alby	713-228-1249	T1	250.00	16-Jun-2018
Kathy W. Smith	615-894-2285	125 Oak Ln, Babs, TN	36123	Leah F. Hahn	615-882-2144	S2	150.00	29-Jan-2019
Paul F. Olowski	615-894-2180	217 Lee Ln., Babs, TN	36123	Leah F. Hahn	615-882-1244	S1	300.00	14-Oct-2018
Myron Orlando	615-222-1672	Box 111, New, TN	36155	Alex B. Alby	713-228-1249	T1	100.00	28-Dec-2018
Amy B. O'Brian	713-442-3381	387 Troll Dr., Fox, KY	25246	John T. Okon	615-123-5589	T2	850.00	22-Sep-2018
James G. Brown	615-297-1228	21 Tye Rd., Nash, TN	37118	Leah F. Hahn	615-882-1244	S1	120.00	25-Mar-2019
George Williams	615-290-2556	155 Maple, Nash, TN	37119	John T. Okon	615-123-5589	S1	250.00	17-Jul-2018
Anne G. Farriss	713-382-7185	2119 Elm, Crew, KY	25432	Alex B. Alby	713-228-1249	T2	100.00	03-Dec-2018
Olette K. Smith	615-297-3809	2782 Main, Nash, TN	37118	John T. Okon	615-123-5589	S2	500.00	14-Mar-2019

C_NAME = Customer name

C_PHONE = Customer phone

C_ADDRESS = Customer address

C_ZIP = Customer zip code

A_NAME = Agent name

A_PHONE = Agent phone

TP = Insurance type

AMT = Insurance policy amount, in thousands of \$

REN = Insurance renewal date

FIGURE 1.8 CONTENTS OF THE AGENT FILE

Database name: Ch01_Text

A_NAME	A_PHONE	A_ADDRESS	ZIP	HIRED	YTD_PAY	YTD_FIT	YTD_FICA	YTD_SLS	DEP
Alex B. Alby	713-228-1249	123 Toll, Nash, TN	37119	01-Nov-2000	26566.24	6641.56	2125.30	132737.75	3
Leah F. Hahn	615-882-1244	334 Main, Fox, KY	25246	23-May-1986	32213.78	8053.44	2577.10	138967.35	0
John T. Okon	615-123-5589	452 Elm, New, TN	36155	15-Jun-2005	23198.29	5799.57	1855.86	127093.45	2

A_NAME = Agent name

A_PHONE = Agent phone

A_ADDRESS = Agent address

ZIP = Agent zip code

HIRED = Agent date of hire

YTD_PAY = Year-to-date pay

YTD_FIT = Year-to-date federal income tax paid

YTD_FICA = Year-to-date Social Security taxes paid

YTD_SLS = Year-to-date sales

DEP = Number of dependents



FIGURE 1.5 EMPLOYEE SKILLS CERTIFICATION IN A POOR DESIGN

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Do we add another column?

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3	06234	Jasmine Patel	Programmer	8/10/2007	Basic Web Design	8/10/2009	Advanced C# programming	8/10/2009	Basic DB manipulation	1/29/2014
4	03373	Franklin Johnson, Jr.	Purchasing Agent	3/15/2004	Advanced Spreadsheets	6/20/2013				
5	13567	Almond, Robert	Analyst	9/30/2014	Basic Process Modeling	9/30/2016	Basic Database Design	5/23/2017		
6	10282	Richardson, Amanda	Clerk	4/11/2013						
7	09382	Susan Mathis	Database Programmer	8/2/2012	Basic DB Design	8/2/2014	Basic Database Manipulation	8/2/2014	Advanced DB Manipulation	5/1/2015
8	14311	Duong, Lee	Programmer	9/1/2016	Basic Web Design	9/1/2018				
9					Master Database Programming					
10					Basic Spreadsheets					
11	09002	Wade Gaither	Clerk	5/20/2012	Advanced Spreadsheets	5/16/2015	Basic Web Design	5/16/2015		
12	13383	Raymond F. Matthews	Programmer	3/12/2014	Basic C# Programming	3/12/2016				
13	09283	Chavez, Juan	Clerk	7/4/2012						
14	04893	Patricia Richards	DBA	6/11/2006	Advanced Database Management	6/11/2008	Advanced Database Manipulation	9/20/2014		
15	13932	Lee, Megan	Programmer	9/29/2015						



Homework/ Break out room

- Consider Figure 1.5
 - Based on the existing format of data shown in Figure 1.5,
 - what will happen with the business manager if he/she want to analyze the data?
 - If the business manager reported that there are 4 persons who work as 'Programmer', do you agree with this analysis?
 - Based on five questions shown in Figure 1.5,
 - what should be improved?
 - To solve problems, design your new data by showing in a tabular form with the explanation.



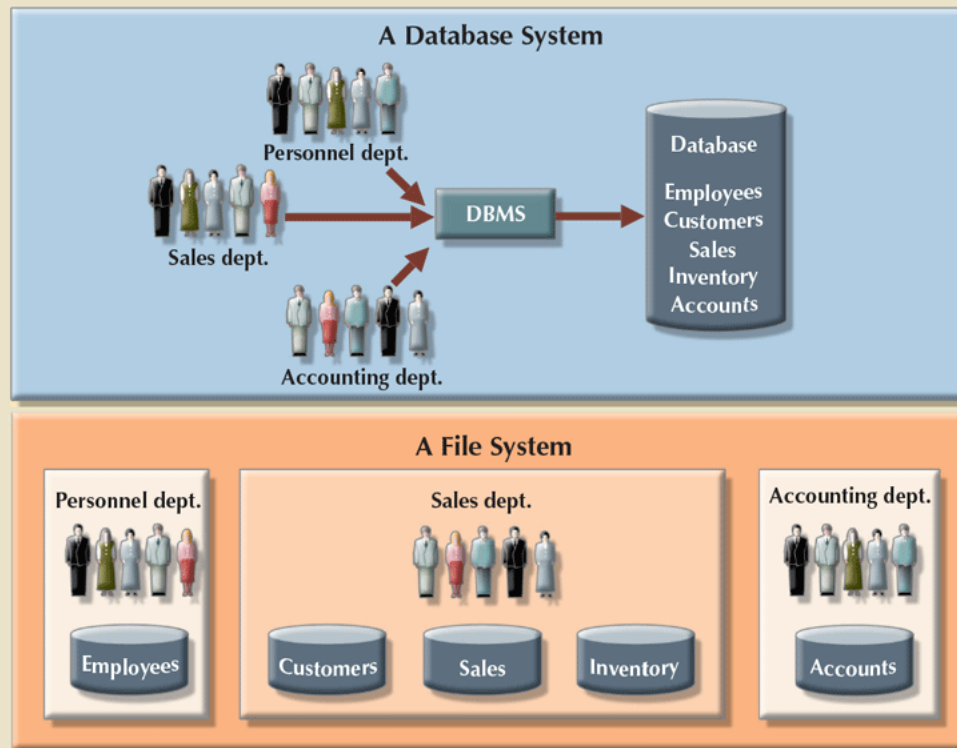
Database Systems (1 of 2)

- Logically related data stored in a single logical data repository
 - Physically distributed among multiple storage facilities
 - DBMS eliminates most of file system's data inconsistency, data anomaly, data dependence, and structural dependence problems
- Current generation DBMS software
 - Stores data structures, relationships between structures, and access paths
 - Defines, stores, and manages all access paths and components



Database Systems (2 of 2)

FIGURE 1.10 CONTRASTING DATABASE AND FILE SYSTEMS





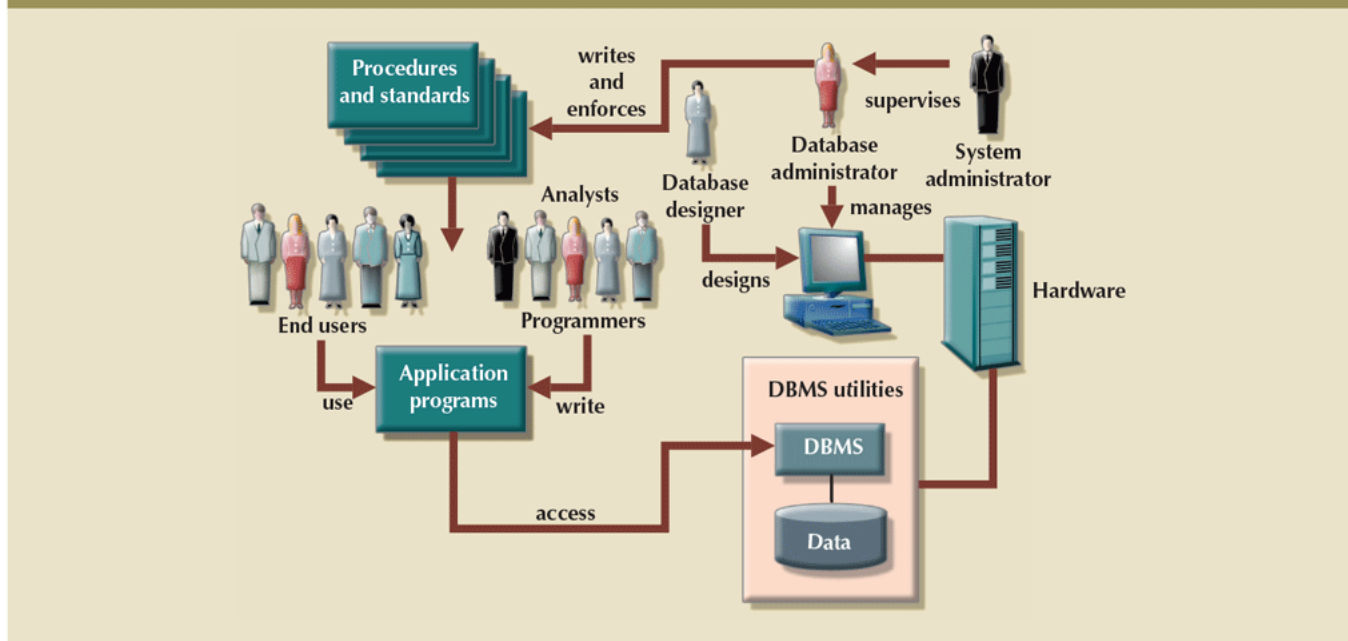
The Database System Environment (1 of 2)

- Database system: organization of components that define and regulate the collection, storage, management, and use of data within a database environment
 - Hardware
 - Software
 - People
 - Procedures
 - Data



The Database System Environment (2 of 2)

FIGURE 1.11 THE DATABASE SYSTEM ENVIRONMENT





DBMS Functions (1 of 3)

- Data dictionary management
 - Data dictionary: stores definitions of data elements and their relationships
- Data storage management
 - Performance tuning ensures efficient performance
- Data transformation and presentation
 - Data is formatted to conform to logical expectations
- Security management
 - Enforces user security and data privacy



DBMS Functions (2 of 3)

- Multiuser access control
 - Sophisticated algorithms ensure that multiple users can access the database concurrently without compromising its integrity
- Backup and recovery management
 - Enables recovery of the database after a failure
- Data integrity management
 - Minimizes redundancy and maximizes consistency



DBMS Functions (3 of 3)

- Database access languages and application programming interfaces
 - Query language: lets the user specify what must be done without having to specify how
 - Structured Query Language (SQL): de facto query language and data access standard supported by the majority of DBMS vendors
- Database communication interfaces
 - Accept end-user requests via multiple, different network environments



Managing the Database System: A Shift in Focus

- Disadvantages of database systems
 - Increased costs
 - Management complexity
 - Maintaining currency
 - Vendor dependence
 - Frequent upgrade/replacement cycles



Preparing for Your Database Professional Career

TABLE 1.3	DATABASE CAREER OPPORTUNITIES	
JOB TITLE	DESCRIPTION	SAMPLE SKILLS REQUIRED
Database Developer	Create and maintain database-based applications	Programming, database fundamentals, SQL
Database Designer	Design and maintain databases	Systems design, database design, SQL
Database Administrator	Manage and maintain DBMS and databases	Database fundamentals, SQL, vendor courses
Database Analyst	Develop databases for decision support reporting	QL, query optimization, data warehouses
Database Architect	Design and implementation of database environments (conceptual, logical, and physical)	DBMS fundamentals, data modeling, SQL, hardware knowledge, etc.
Database Consultant	Help companies leverage database technologies to improve business processes and achieve specific goals	Database fundamentals, data modeling, database design, SQL, DBMS, hardware, vendor-specific technologies, etc.
Database Security Officer	Implement security policies for data administration	DBMS fundamentals, database administration, SQL, data security technologies, etc.
Cloud Computing Data Architect	Design and implement the infrastructure for next-generation cloud database systems	Internet technologies, cloud storage technologies, data security, performance tuning, large databases, etc.
Data Scientist	Analyze large amounts of varied data to generate insights, relationships, and predictable behaviors	Data analysis, statistics, advanced mathematics, SQL, programming, data mining, machine learning, data visualization



Summary

- Data consists of raw facts and is usually stored in a database
 - Database design defines the database structure
 - Can be classified according to the number of users, location, as well as data usage and structure
 - Databases evolved from manual and computerized file systems
 - There are some limitations of file system data management
 - DBMSs were developed to address the file system's inherent weaknesses